University of Eloued
Faculty of Exact Sciences
Department of Computer science
For Second-year Master: IA & Data Science

University season: 2026/2027
Module:Big Data Analytics

TP04: Containerization and Big Data Environment Deployment Using Docker

Submission rule: All work must be uploaded to the student's GitHub account. Include source .files, screenshots, and a README.md

• Educational Objectives

- Understand Docker images, containers, volumes, networks, and port mapping.
- Use basic Docker commands to download, run, inspect, start, and stop containers.
- Build a custom image for a Python data-processing application using a Dockerfile.
- Use Docker Compose to define and run an application environment.

• Prerequisites and Tools

- Docker Desktop on Windows, or Docker Engine on Linux.
- Python basics and familiarity with CSV files.
- A CSV dataset previously used in TP02 or TP03.

• Guided Lab Tasks

Task 1 — Installation

Verify Docker Installation and Create an account on Docker Hub.

Task 2 — Image

- Create an image for dataset Lab 3 (TP3).
- Push this image to Docker Hub.
- Pull the image that was uploaded to Docker Hub.

Task 3 — Containers

- Create three containers from the downloaded image.
- Create a network to connect the three containers.
- Create and launch the three containers (the `-p` flags allow for port mapping between the host machine and the containers).
- Verify that the three containers are running correctly by executing the appropriate command.

Task 4 — Run the Application with Docker Compose

Create a file named compose.yaml:

```
services:
  data-analysis:
    build: .
    container_name: tp04-analysis
    volumes:
      - ./data:/app/data:ro
    restart: "no"
```

Execute the following commands:

```
docker compose up --build
docker compose ps -a
docker compose down
```

Questions:

- What problem does Docker Compose solve ?
- Compare docker run with docker compose up ?
- How could you extend the Compose project to include a database service ?

Task 5 — Report summarizing

Write a brief report summarizing the answers to the following questions:

- What is Docker, and why is it useful in data-science projects ?
- Explain the difference between a Docker image and a container ?
- Explain the purpose of FROM, WORKDIR, COPY, RUN, and CMD in the Dockerfile ?

Good luck